GEOMETRIC GRAPH RESEARCH / SECOND DEEP INVESTIGATION

From complete geometric graphs to local Poisson lines

Adversarial geometry, exact intensity fields, sampling bias, and predictions without fitted exponents

16 September 2026

Main result. In a fixed interior neighborhood of a dense complete graph on independent random cities, the appropriate magnification is of order n^2, not the $n^{1/2}$ scale of city spacing. The local segment arrangement has a Poisson-line limit whose angular intensity can be computed directly from the sampling density. Classical results for that limiting tessellation then supply testable distributions and exact reference constants without regression.

An important falsification. The earlier predictor $\widehat{A} = 3.709513\, r\sqrt{d_2 d_3}$ remains an average-case heuristic, not a universal geometric relation. A fixed nine-city family makes its multiplicative error diverge; one verified example underestimates area by a factor of 330,617.

New numerical work. This note analyzes 16,000 independent complete-graph probes, with city counts up to 1,024; 15,997 target cells are geometrically certified. It also includes 10,000 independent Poisson-line reference cells, 67 successful comparisons against whole-graph polygonization, deterministic integral checks, and 41 adversarial examples. No cell-area model is trained in this phase.

Status. The Poisson-tessellation identities used here are classical and attributed. The local complete-graph connection is developed through a proof argument and direct tests in this note, not presented as an independently peer-reviewed result or a verified novelty claim. Finite-n deviations and the limits of the convergence argument are stated explicitly.

Reproducibility. The companion archive contains source code, graph seeds, per-graph outputs, integral checks, figures, and this document's LaTeX source. Earlier large archives are not duplicated.

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1 What changes from the earlier investigation

The original experiment studied the bounded polygonal faces produced by all straight segments joining seven cities. Successive regressions related face area to graph-edge density, crossings, inradius, and nearest crossing distances. Those experiments found useful predictors, but also exposed dependence on the definition of locality. This investigation separates three questions that should not be conflated:

1. Can a deterministic formula approximate *every* face? The nearest-crossing formula fails this test.
2. Can one derive the *mean local geometry* from the distribution of the cities? Yes: an explicit angular segment field yields exact finite-n expectations for length and crossings.
3. Does an entire local random-cell distribution emerge at high density? A rare-event argument identifies a Poisson-line limit in the bulk, and independent simulations support its parameter-free predictions.

A fitted exponent is an attribute of a chosen regression and sampling procedure. A limit distribution, with stated hypotheses, can instead predict several distinct observables simultaneously. That is the organizing principle here.

1.1 Notation and sampling convention

Let $X_1, \dots, X_n$ be independent cities in the plane. Draw the segment $[X_i, X_j]$ for every $i < j$. Fix a spatial probe x independently of the graph, strictly inside the sampling domain. Write $C_n(x)$ for its containing face when that face is bounded. The following quantities are different:

Symbol	Meaning
A, P, S	Area, perimeter, and side count of a polygonal face.
r	Maximum inscribed-circle radius; its center may differ from x.
$\delta_n(x)$	Distance from the fixed probe x to the nearest *finite graph segment*.
d_j	Distance from the face's area centroid to the jth nearest proper graph crossing, used only in the earlier predictor and its adversary.
$\lambda_x(\theta)$	Direction-resolved segment-intensity coefficient, defined below.
$L(x), v(x)$	Total length-intensity and crossing-intensity coefficients.
C_0	Zero/Crofton cell: the cell containing a fixed point in a stationary line tessellation.
C_{typ}	Count-selected typical cell in that stationary tessellation.

A fixed probe is *not* a face centroid or an incenter. A randomly selected spatial point and a randomly selected face induce different distributions. This distinction becomes quantitatively important.

2 A nine-city adversary: crossings can belong to the wrong face

For $0 < \varepsilon \le 10^{-2}$, take the nine cities

$$
\begin{aligned}
&(-1, 0), \quad (1, 0), \quad (0, \varepsilon),\\
&(-1.07\varepsilon, 2.09\varepsilon), \quad (-0.73\varepsilon, 3.23\varepsilon),\\
&(0.83\varepsilon, 3.07\varepsilon), \quad (1.19\varepsilon, 1.93\varepsilon),\\
&(-0.81, 2.11), \quad (0.61, 3.13).
\end{aligned}
$$

The target triangle has the first three vertices. It remains an intact bounded face of the complete graph: the remaining cities lie in the convex region

$$y \geq \varepsilon(1 + |x|),$$

above its two sloping supporting lines, so their interconnecting segments cannot enter the triangle. Segments from the two base vertices to those cities also leave on the exterior side.

The target face has

$$A = \varepsilon, \qquad r = \frac{\varepsilon}{1 + \sqrt{1 + \varepsilon^2}} \sim \frac{\varepsilon}{2}.$$

The apex and four nearby cities form a small convex pentagon. Its internal crossings are outside the target face but at distance $O(\varepsilon)$ from the target centroid. Consequently $d_2, d_3 = O(\varepsilon)$ and

$$\widehat{A} = 3.7095130346\, r\sqrt{d_2 d_3} = O(\varepsilon^2), \qquad \boxed{\frac{A}{\widehat{A}} \longrightarrow \infty.}$$

Thus the predictor has no universal finite multiplicative error bound, even for triangular faces of a fixed K_9.

Triangle height ε	Actual / predicted area
10^{-2}	33.685
10^{-3}	331.239
10^{-4}	3,306.788
10^{-5}	33,062.279
10^{-6}	330,617.195

Across the 41 computed heights, the hull area is 4.5312, there are 64 proper crossings and 92 bounded faces, and the target triangle's symmetric difference with the polygonized face is zero at the reported floating-point precision. The analytic order argument, not those finite checks alone, establishes divergence.

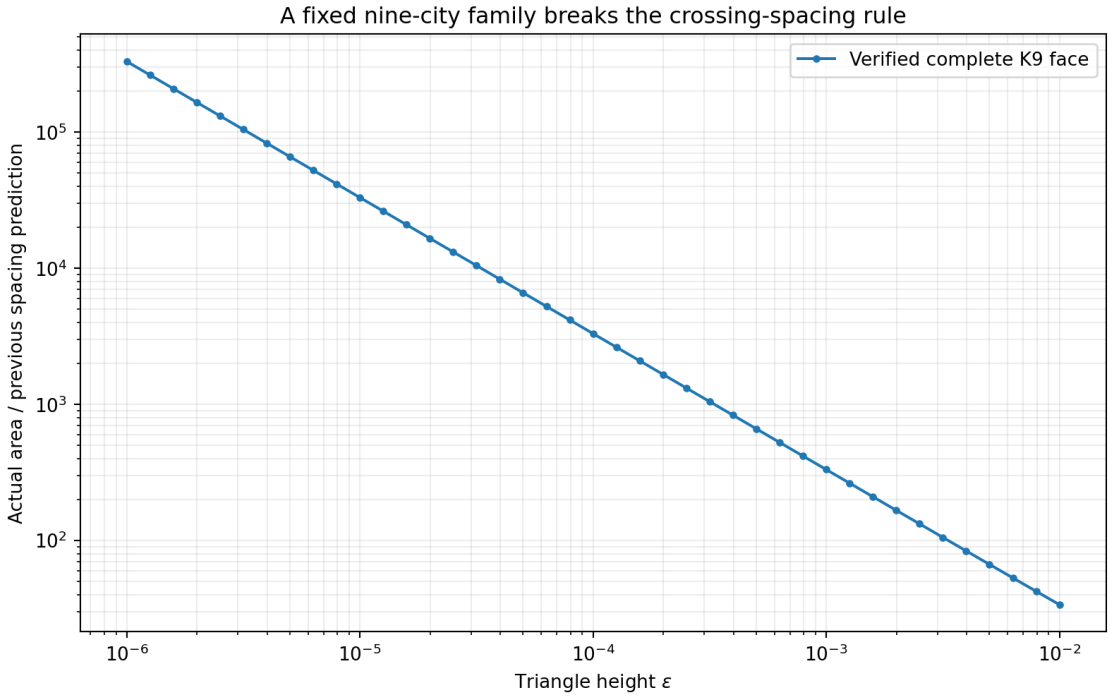


Figure 1: The earlier spacing heuristic fails without bound on a controlled family. Crossings near a face need not constrain that face's long direction.

This does not erase the heuristic's earlier average-case success. It identifies its missing topological information: the closest crossings can be vertices of entirely different cells. Enlarging a fixed list of nearest crossings cannot generally fix the issue, because an unrelated nearby cluster can supply any prescribed finite number of them.

3 Exact local intensity fields from the city distribution

3.1 Direction-resolved segment intensity

Let the city density be f, and $u_\theta = (\cos\theta, \sin\theta)$ for $0 \le \theta < \pi$. Define

$$\boxed{\lambda_x(\theta) = \int_0^\infty \int_0^\infty (a+b) f(x + a u_\theta) f(x - b u_\theta)\, \mathrm{d}a\, \mathrm{d}b.} \tag{1}$$

The integral selects pairs of endpoints on opposite sides of x along a line. The factor $a + b$ is the planar line-coordinate Jacobian. Unlike a local crossing count, this is a deterministic function of the underlying endpoint distribution.

For uniform sampling from a convex domain D of area S_D, let $a_\theta(x)$ and $b_\theta(x)$ be the distances from x to the domain boundary in the two opposite directions. Direct integration gives

$$\boxed{\lambda_x(\theta) = \frac{a_\theta(x) b_\theta(x)[a_\theta(x) + b_\theta(x)]}{2 S_D^2}.} \tag{2}$$

Both the amount of available chord and the endpoint geometry matter.

Define

$$L(x) = \int_0^\pi \lambda_x(\theta)\,\mathrm{d}\theta, \tag{3}$$

$$v(x) = \frac{1}{2}\int_0^\pi\int_0^\pi \lambda_x(\theta)\lambda_x(\phi)|\sin(\theta-\phi)|\,\mathrm{d}\theta\,\mathrm{d}\phi. \tag{4}$$

The sine factor is the intersection Jacobian: parallel directions generate no crossings; transverse directions do.

Proposition 1 (Exact first-moment identities)**.** *For an integrable planar set B, the complete graph on n independent cities satisfies*

$$\mathbb{E}[\textit{total segment length in } B] = n(n-1)\int_B L(x)\,\mathrm{d}x, \tag{5}$$

$$\mathbb{E}[\textit{number of proper crossings in } B] = (n)_4\int_B v(x)\,\mathrm{d}x, \tag{6}$$

where $(n)_4 = n(n-1)(n-2)(n-3)$. Shared city endpoints are excluded from the proper-crossing count.

Proof. Parameterize an ordered pair of endpoints by its line angle, signed normal offset, and the two positions along that line. The Jacobian is the endpoint separation. Restricting a point of the segment to lie in B gives twice (1) for a single unordered segment's expected length density. Multiplication by $\binom{n}{2}$ gives (5).

For two independent segments the corresponding angular length measures are $2\lambda_x(\theta)\,\mathrm{d}\theta$ and $2\lambda_x(\phi)\,\mathrm{d}\phi$. Transforming their two normal offsets to their intersection point contributes $|\sin(\theta-\phi)|$. The resulting crossing density is $8v(x)$. There are $3\binom{n}{4} = (n)_4/8$ unordered pairs of edges with four distinct endpoints, giving (6). Pairs sharing a city contribute only that excluded endpoint, almost surely. □

These are exact expectation formulas for segments and crossings at finite n. They are *not* claims that the finite-n face process is Poisson, nor that its exact face-count density is $(n)_4 v$.

3.2 Checks and special cases

For the centered unit square $[-1/2, 1/2]^2$,

$$\lambda_0(\theta) = \frac{1}{8\max(|\cos\theta|, |\sin\theta|)^3},$$

$$\boxed{L(0) = \frac{\sqrt{2}+\log(1+\sqrt{2})}{4}, \qquad v(0) = \frac{5}{48}.} \tag{7}$$

The square-wide crossing coefficient is $\int_D v = 25/864$. Thus the local expected crossing density at the center is exactly 3.6 times its spatial average. This result comes from endpoint integration, not a fitted heatmap.

The unit-radius disk gives, at its center,

$$\lambda_0(\theta) = \frac{1}{\pi^2}, \qquad L(0) = \frac{1}{\pi}, \qquad v(0) = \frac{1}{\pi^3}.$$

At $x = (s, 0)$ in that disk,

$$\lambda_s(\theta) = \frac{(1-s^2)\sqrt{1-s^2\sin^2\theta}}{\pi^2}, \qquad L(s) = \frac{2(1-s^2)E(s^2)}{\pi^2},$$

where $E(m) = \int_0^{\pi/2} \sqrt{1 - m \sin^2 t}\, \mathrm{d}t$ is the complete elliptic integral of the second kind.

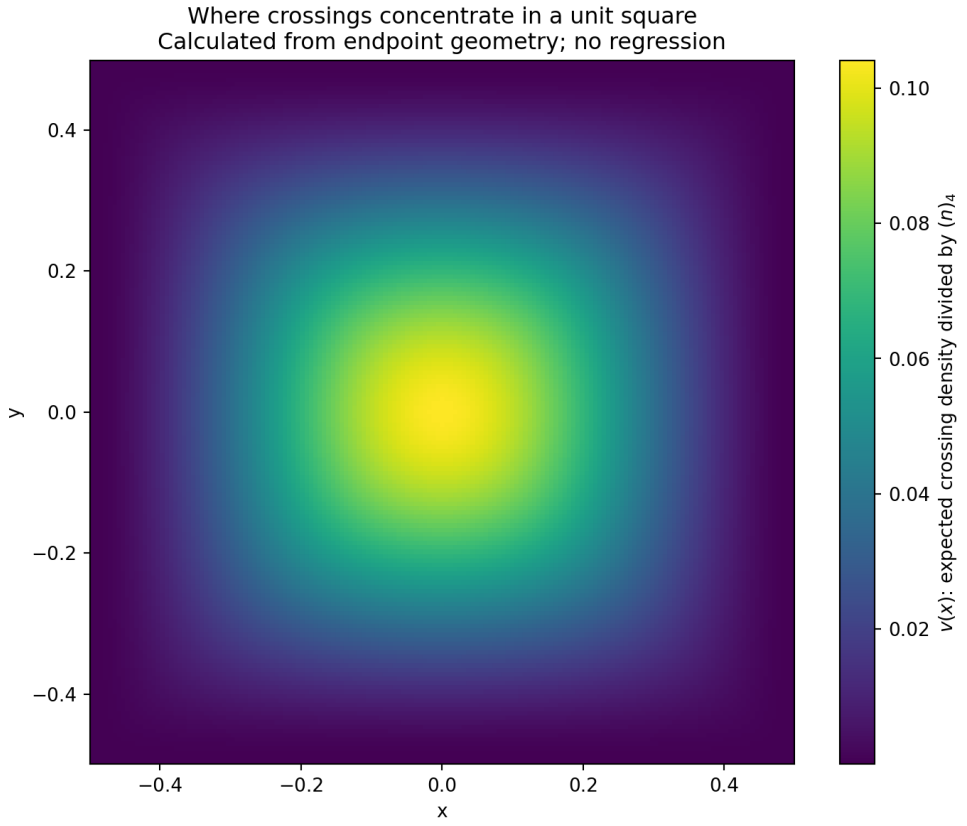


Figure 2: Deterministic quadrature of the exact square crossing field. The original graph can be highly irregular, but its expectation has a calculable spatial structure.

Global consistency checks are

$$\int_{\text{square}} L = \frac{2 + \sqrt{2} + 5\log(1+\sqrt{2})}{30}, \qquad \int_{\text{square}} v = \frac{25}{864},$$
$$\int_{\text{disk}} L = \frac{64}{45\pi}, \qquad \int_{\text{disk}} v = \frac{1 - 35/(12\pi^2)}{24}.$$

The first pair matches the previous random-square crossing/face-count calculation. The saved quadrature checks refine angular grids from 128 to 1,024 points and reproduce the known global values to approximately 10^{-8} or better at the finest settings; this numerical agreement is a validation, not a substitute for the identities.

4 A local Poisson-line limit

4.1 The correct magnification

Independent planar cities have ordinary neighbor distances of order $n^{-1/2}$ in the bulk. Their complete graph, however, contains order n^2 long segments. A disk of radius ε is hit by any one generic segment

with probability of order ε, not ε^2. To see a nontrivial number of cuts requires

$$n^2\varepsilon \asymp 1, \quad \text{hence} \quad \boxed{\varepsilon \asymp n^{-2}.}$$

The old neighborhood calibration based on city spacing therefore mixed two distinct scales. A radius of order $n^{-1/2}$ is of order $n^{3/2}$ *cell widths*; it grows less local relative to the interior tessellation as n increases.

4.2 Proposition and proof argument

The following statement is developed here with a rare-event proof argument. Its method is classical Poisson approximation for locally dependent indicators [1]; related point-process limits for geometric functionals are discussed by Schulte and Thäle [2]. Neither citation is being presented as a publication of this exact complete-graph statement.

Proposition 2 (Bulk local segment limit)**.** *Let cities be independent and uniform in a compact convex planar domain with nonempty interior, and fix an interior point x. Under the spatial magnification*

$$y \longmapsto n^2(y - x),$$

the segment traces in bounded windows converge in distribution to a stationary Poisson line process with line-parameter intensity

$$\lambda_x(\theta)\,\mathrm{d}p\,\mathrm{d}\theta, \tag{8}$$

where p is signed normal offset and $\theta \in [0, \pi)$ is line direction. The cell containing the probe converges in distribution to the zero cell of this process.

The same argument extends to bounded, compactly supported densities for which the transverse line-intensity limit is continuous at the chosen probe and nondegenerate. Positivity in a neighborhood of the probe provides the needed local directional nondegeneracy. No boundary-uniform claim is made.

Proof argument. Fix a bounded magnified window and let $\varepsilon = n^{-2}$. For each pair $i < j$, record its line parameters when its finite segment meets the physical window $x + \varepsilon W$. Such a pair contributes a rare marked event of probability $O(\varepsilon)$.

Mean measure. The line-coordinate change of variables used in Proposition 1, with normal offset $p\varepsilon$, gives the limiting mean measure (8). For a uniform convex domain, chord endpoint lengths vary continuously under small transverse shifts except on a negligible angular set, and boundedness gives dominated convergence.

Dependence. Pairs of edges with four distinct city labels are independent. Only edges sharing a label require control. Conditional on a shared city a distance d from x, the other endpoint must lie in a thin cone to make its segment hit an $O(\varepsilon)$ disk. Under bounded density and compact support,

$$\mathbb{P}(\text{hit} \mid X_i = z) \le C \min(1, \varepsilon/|z - x|).$$

Integrating its square in polar coordinates yields

$$\mathbb{P}(\text{two edges sharing a city both hit}) \le C\varepsilon^2[1 + \log(1/\varepsilon)].$$

There are $O(n^3)$ shared-label pairs. Thus the aggregate dependency contribution is

$$O\Big(n^3\varepsilon^2 \log(1/\varepsilon)\Big) = O\Big(\frac{\log n}{n}\Big) \longrightarrow 0.$$

The sum of products of rare-event probabilities over those dependency neighborhoods is $O(n^3\varepsilon^2) = O(n^{-1})$. The multivariate Chen–Stein argument applied to disjoint line-parameter bins therefore gives asymptotically independent Poisson counts. Tightness of the finite mean measures on compact line windows promotes these finite-bin limits to local point-process convergence. The displayed rate controls the dependency term, not an asserted uniform rate for every geometric observable.

Segments become lines locally. The probability of any city lying in the physical probe window is $O(n\varepsilon^2) = O(n^{-3})$. Except on this event, an intersecting segment has no endpoint in the window and traces an entire line across it.

The containing cell. The nondegenerate limiting directional measure almost surely encloses the origin in a bounded polygon, with no triple intersections. First restrict to a large magnified window on the event that its boundaries do not truncate the zero cell. Polygonal intersection is then continuous at generic configurations. Letting that window grow proves the asserted local-cell distributional convergence. An original-city endpoint in the window has already been excluded by a probability tending to zero. □

Scope of the proposition. This argument concerns fixed bulk windows and distributional convergence. It does not by itself establish convergence of unbounded moments such as area and perimeter: that would need a uniform-integrability argument, not supplied here. Consequently the exact limiting means below are reference predictions, and the finite-n mean comparisons are evidence rather than an additional moment-convergence theorem. Independent proof review would also be appropriate before treating this research note as a formal publication.

5 Predictions from the limiting process

The Poisson-line statements in this section are classical or are short direct consequences of the classical model. References include Calka [3], Hilhorst–Calka [4], and Schneider [5]. Their use here is to test the complete-graph connection, not to claim those identities as new discoveries.

5.1 A full containment law

Let a stationary line process have intensity $\lambda(\theta)\,\mathrm{d}p\,\mathrm{d}\theta$, and write $L = \int \lambda$. For a compact convex set K containing the origin, let $w_K(v_\theta)$ be its width in the normal direction $v_\theta = (-\sin\theta, \cos\theta)$. The number of lines meeting the interior of K is Poisson with mean $\int \lambda(\theta) w_K(v_\theta)\,\mathrm{d}\theta$. Therefore

$$\boxed{\mathbb{P}(K \subset C_0) = \exp\left[-\int_0^\pi \lambda(\theta) w_K(v_\theta)\,\mathrm{d}\theta\right].} \tag{9}$$

This specifies much more than one fitted moment. In the isotropic case $\lambda = c$, Cauchy's perimeter formula gives

$$\mathbb{P}(K \subset C_0) = e^{-cP(K)}.$$

At line-length intensity one, $c = 1/\pi$, so the exponent is $-P(K)/\pi$. Convex shapes containing the probe with the same perimeter have the same containment probability in this isotropic model. This is not a claim that their probabilities of being the *exact cell shape* are the same.

5.2 Nearest-edge clearance has an exponential limit

A disk of radius R is met by a mean of $2LR$ lines, whatever the angular distribution. Thus its void probability is e^{-2LR}. Define the exact finite-n local length scale

$$\ell_n(x) = n(n-1)L(x).$$

The local-limit proposition yields

$$\boxed{2\ell_n(x)\delta_n(x) \;\Rightarrow\; \mathrm{Exp}(1).} \tag{10}$$

At the center of the unit-radius disk this is

$$T_n = \frac{2n(n-1)}{\pi}\delta_n(0) \;\Rightarrow\; \mathrm{Exp}(1).$$

There is no estimated exponent or fitted rate. This is clearance of the *fixed probe*, not the maximum inscribed radius of its cell.

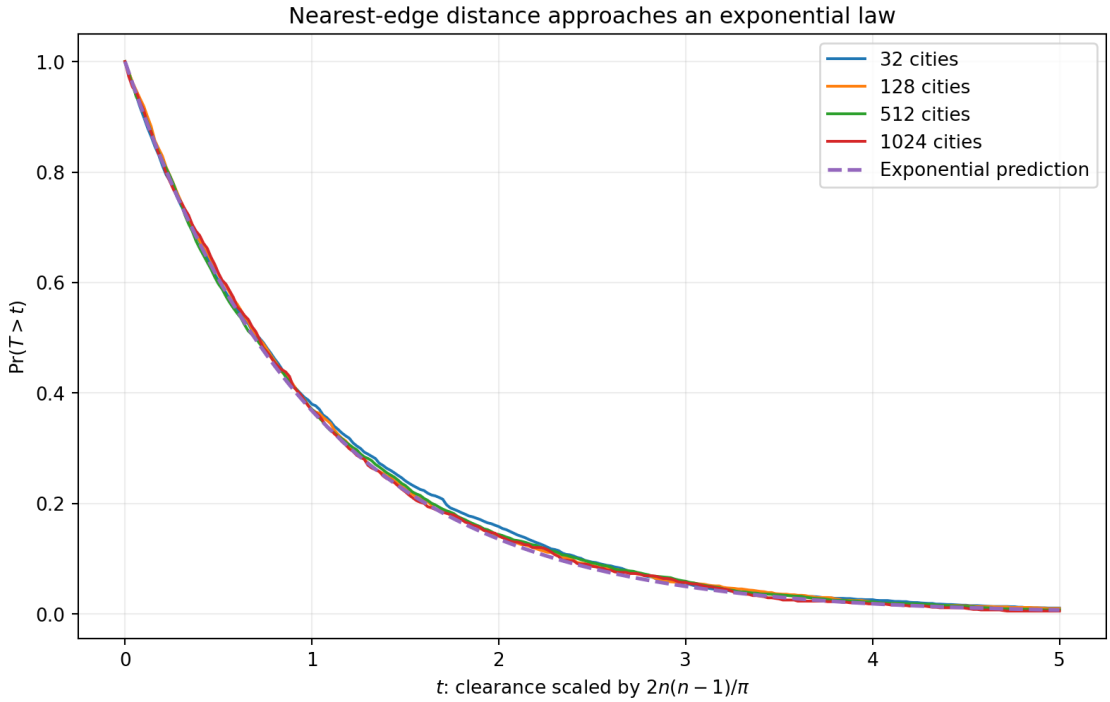


Figure 3: Survival functions of scaled finite-segment clearance, compared with e^{-t}. Each observed clearance comes from a different independently generated complete graph.

5.3 Isotropic zero-cell constants

For isotropic line intensity $c\,\mathrm{d}p\,\mathrm{d}\theta$, a segment from the origin to y is crossed by a mean of $2c|y|$ lines. Hence

$$\mathbb{P}(y \in C_0) = e^{-2c|y|}, \qquad \mathbb{E}A_0 = \int_{\mathbb{R}^2} e^{-2c|y|}\,\mathrm{d}y = \frac{\pi}{2c^2}.$$

The line-length intensity is $L = \pi c$, and the crossing/cell intensity is $v = \pi c^2$. The typical-cell mean area is $1/v$. Campbell's formula for candidate vertices of the zero cell gives

$$\mathbb{E}S_0 = v\mathbb{E}A_0 = \frac{\pi^2}{2}.$$

Integrating the lengths of candidate boundary lines similarly gives $\mathbb{E}P_0 = L\mathbb{E}A_0$.

In coordinates rescaled to have $L = 1$,

Quantity	Count-selected typical cell	Point-selected zero cell
Mean area	π	$\pi^3/2$
Mean perimeter	2π	$\pi^3/2$
Mean number of sides	4	$\pi^2/2$
Triangle probability	$2 - \pi^2/6$	$(25 - 36\log 2)\pi^2/6$

The triangle probabilities, due to classical work summarized in [4], are approximately 0.355066 and 0.076821 respectively. The identical numeric zero-cell area and perimeter means at $L = 1$ have different physical dimensions; it is a normalization coincidence.

5.4 A conditional perimeter distribution, not just a mean

Calka's conditional Gamma law [3, Corollaries 2–3], converted to $L = 1$ coordinates, states

$$\boxed{\frac{P_0}{\pi}\,\middle|\, S_0 = m \sim \text{Gamma}(m, \text{rate } 1).} \tag{11}$$

For the count-selected typical cell the shape parameter is instead $m - 2$. The two extra Gamma degrees reflect the area weighting, which is homogeneous of degree two.

This permits a no-fitting distributional test. For each measured rescaled complete-graph cell, compute

$$U = F_{\text{Gamma}(S,1)}(P/\pi).$$

The limiting law makes U uniform on $[0, 1]$. Neither side count nor perimeter is adjusted to make the test fit.

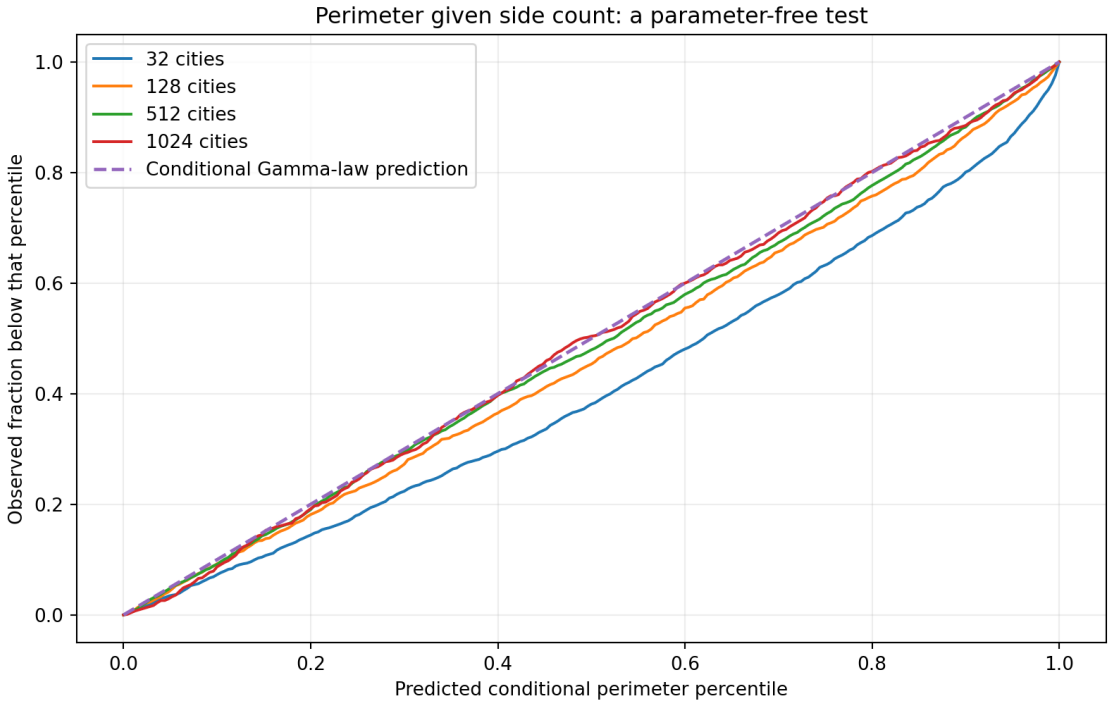


Figure 4: Observed probability-integral-transform curves approach the diagonal. The discrepancy is not zero at finite n, and formal tests still detect it at $n = 512$.

6 Sampling bias and a connection to convex duality

6.1 Why four sides and 4.935 sides can both be correct

In a stationary tessellation, spatial point sampling weights cells in proportion to their areas. For a translation-invariant observable g,

$$\mathbb{E}g(C_0) = \frac{\mathbb{E}[A(C_{\text{typ}})g(C_{\text{typ}})]}{\mathbb{E}A(C_{\text{typ}})}. \tag{12}$$

This classical identity [5] is a sampling statement, not a contradiction with Euler's formula. In particular,

$$\mathbb{E}A_0 = \frac{\mathbb{E}A_{\text{typ}}^2}{\mathbb{E}A_{\text{typ}}},$$

so a spatial point systematically sees larger cells.

For isotropic Poisson lines,

$$\boxed{\frac{\mathbb{E}A_0}{\mathbb{E}A_{\text{typ}}} = \frac{\pi^2}{2} \approx 4.9348.}$$

The same number is the expected side count of the zero cell; this is not the product of two independently selected averages. A point-picked cell is almost five times as large *on average* as a count-picked cell in this homogeneous limiting model. The ratio is not a statement about the finite, inhomogeneous original seven-city graph.

6.2 Angular information becomes a deterministic convex body

Let v_θ again denote the unit normal. Define the centrally symmetric convex body Z by its support function

$$h_Z(y) = \frac{1}{2}\int_0^\pi \lambda(\theta)|y \cdot v_\theta| \, \mathrm{d}\theta. \tag{13}$$

Equivalently, Z is a continuous Minkowski sum of symmetric normal-direction segments, a *zonoid*. Define its polar $Z^\circ = \{y : h_Z(y) \le 1\}$.

The separating-line formula gives

$$\mathbb{P}(y \in C_0) = e^{-2h_Z(y)}.$$

Integration of the level sets of the homogeneous function h_Z gives

$$\mathbb{E}A_0 = \frac{1}{2}\operatorname{area}(Z^\circ).$$

The area of the zonoid is exactly the crossing intensity,

$$\operatorname{area}(Z) = v.$$

Combining these identities with the vertex calculation yields

$$\boxed{\mathbb{E}S_0 = \frac{\mathbb{E}A_0}{\mathbb{E}A_{\mathrm{typ}}} = \frac{1}{2}\operatorname{area}(Z)\operatorname{area}(Z^\circ).} \tag{14}$$

This is a classical connection between random tessellations and the deterministic convex volume product; see [5]. It is derived here to explain its relevance to complete-graph local geometry.

The centrally symmetric planar volume-product inequalities give

$$8 \le \operatorname{area}(Z)\operatorname{area}(Z^\circ) \le \pi^2,$$

with parallelogram and ellipse extremizers. Consequently

$$\boxed{4 \le \mathbb{E}S_0 = \frac{\mathbb{E}A_0}{\mathbb{E}A_{\mathrm{typ}}} \le \frac{\pi^2}{2}.} \tag{15}$$

Schneider's theorem establishes the corresponding bounds more generally for weighted faces [5]; the planar lower bound also has a direct proof in [6]. The left endpoint corresponds to two nonparallel line directions, and the isotropic process attains the right endpoint. Affine images of the isotropic process also attain the upper value.

This explains why $\pi^2/2$ is not an attractive fitted number: it is an extremal convex-geometric constant in the correct sampling law.

7 Direct complete-graph experiments and numerical safeguards

7.1 Design

All new random-city experiments use genuine independent sampling, with *no minimum-distance rejection filter*. Earlier experiments sometimes rejected near-coincident cities; those samples are not silently reused

for this limit test. The unit disk is sampled with independent uniform angle and square-root radial coordinate. The square is $[-1/2, 1/2]^2$.

Every measurement is from an independently generated complete graph. The graph is probed at the center or at the fixed off-center point $(0.65, 0)$. Lengths are rescaled by $\ell_n = n(n-1)L(x)$, areas by ℓ_n^2. There is one target cell per graph, so bootstrap resampling is at the graph level.

Domain / probe	Cities	Independent graphs	Certified cells	Mean scaled clearance T
Disk / center	32	2,500	2,500	1.0460
Disk / center	128	2,000	2,000	1.0338
Disk / center	256	2,000	2,000	0.9890
Disk / center	512	3,000	3,000	1.0202
Disk / center	1,024	1,000	1,000	1.0086
Disk / $(0.65, 0)$	128	2,000	1,997	1.1300
Disk / $(0.65, 0)$	256	1,500	1,500	1.0644
Square / center	128	2,000	2,000	1.0122
Total		16,000	15,997	

Three off-center cells have a city inside the expanding local probe window and are explicitly marked uncertified. Their segment-clearance and segment-count measurements are still retained, while their unverified cell moments are not used. These exclusions are visible in the raw files. This can induce a small conditioning effect in those specific off-center moment estimates; no claim of zero bias is made.

7.2 Computing one cell without constructing billions of faces

Each graph's $\binom{n}{2}$ finite segments is inspected. Their true distances to the probe provide clearance and disk-hit counts. An adaptive square around the probe is then clipped by the supporting halfplanes of segments that actually cross the square. If no city lies inside the square, each crossing segment spans the window as a line. The result is accepted only when the candidate cell lies strictly inside the window. Otherwise the window is expanded, or the case is logged as uncertified. At $n \le 64$, whole-graph polygonization supplies a fallback.

The absence of endpoints and the strict window containment are important: extending every supporting line globally would generally create the wrong finite-graph cells. Here the local extension is used only where it is valid.

For $n = 1{,}024$ each graph has 523,776 edges. The algorithm does *not* enumerate its potentially billions of bounded faces. It constructs only the probe cell. Reproducible per-graph random seeds are stored, allowing reconstruction of all original city coordinates.

Independent small-graph checks compared local cells against complete Shapely polygonization. Of 90 attempted small-graph cases, 67 certified local cells were compared; the maximum relative symmetric-difference error was 1.14×10^{-14}. The remaining local cases were not counted as successful geometry checks. Inradii are computed by convex supporting-line optimization, not by a whole-drawing fixed-tolerance search.

7.3 Results at the disk center

Cities	Mean sides	95% graph-bootstrap interval	Triangle fraction	Mean rescaled area
32	4.7340	[4.6984, 4.7728]	0.0604	18.3855
128	4.8860	[4.8385, 4.9385]	0.0745	17.1052
256	4.8775	[4.8320, 4.9260]	0.0790	16.1054
512	4.9257	[4.8847, 4.9670]	0.0770	16.0976
1,024	4.9190	[4.8530, 4.9900]	0.0780	15.6365
Poisson reference	4.9348	exact model mean	0.076821	15.5031

The last row contains exact limiting-model values, not fitted targets. The observations are consistent with approach to that model, with sampling noise and visible finite-size bias. For example, the $n = 512$ perimeter mean has a confidence interval that remains above the exact limiting mean; matching one statistic does not establish exact finite-n equality.

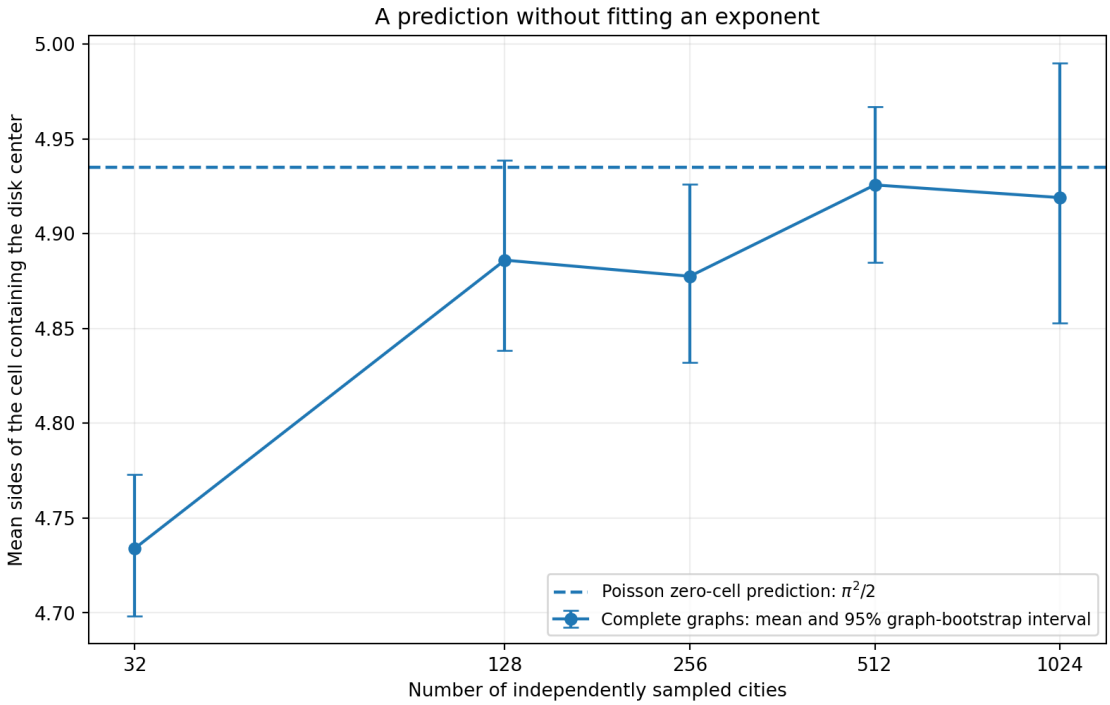


Figure 5: Point-selected side count is predicted to approach $\pi^2/2$, not the count-selected mean four. The nonmonotone sample means at 128 and 256 cities are retained.

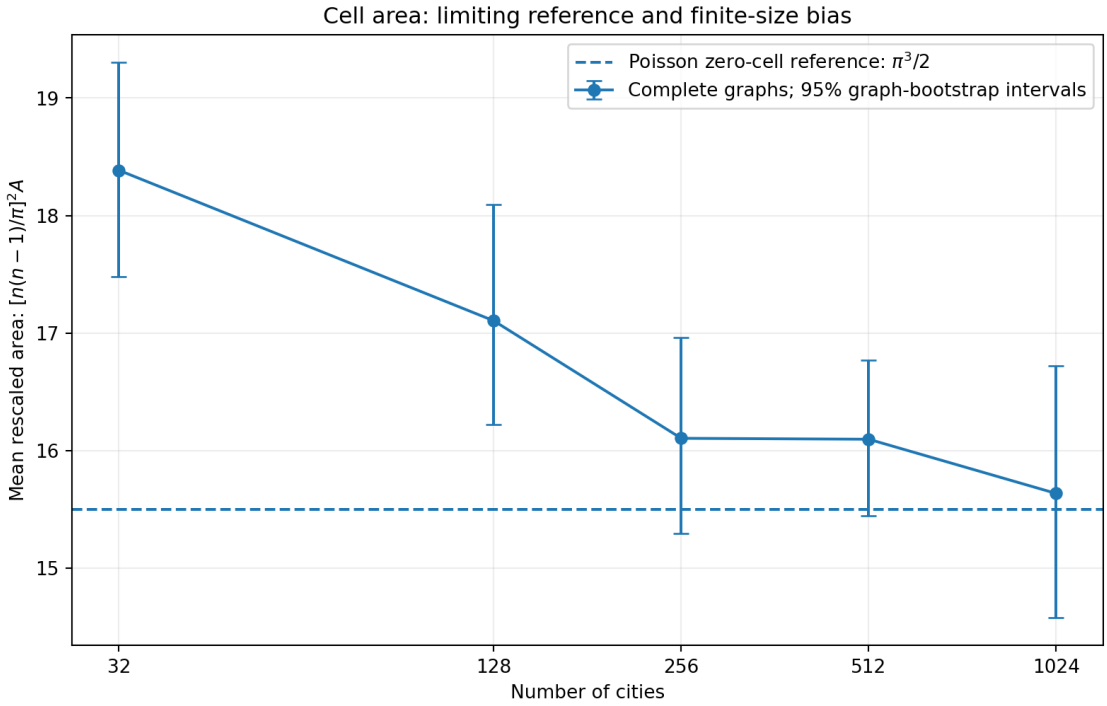


Figure 6: Rescaled-area sample means and graph-bootstrap uncertainty. Distributional convergence alone is not a proof of convergence of these unbounded means.

7.4 Full distribution tests and finite-size discrepancies

For the conditional Gamma transform in (11), the Kolmogorov–Smirnov discrepancies from a uniform distribution are:

Cities	Independent graphs	KS discrepancy	p-value against the exact limit law
32	2,500	0.12273	2.8×10^{-33}
128	2,000	0.05006	8.5×10^{-5}
256	2,000	0.03492	0.0149
512	3,000	0.03061	0.0071
1,024	1,000	0.02038	0.793

The finite-graph distribution still differs detectably from the limiting Gamma law at $n = 512$. At $n = 1{,}024$ that test does not reject it, but non-rejection is not proof of equality. The decreasing discrepancy is the relevant evidence for convergence; the tests are not a model-selection exercise.

A separate 10,000-cell reference sample was generated from an actual isotropic Poisson line process of length intensity one. It is not labeled as complete-graph data. All reference cells passed the generous-window enclosure check. It gives mean side count 4.9363 and mean area 15.3254, illustrating ordinary Monte Carlo variation even when the exact model is simulated.

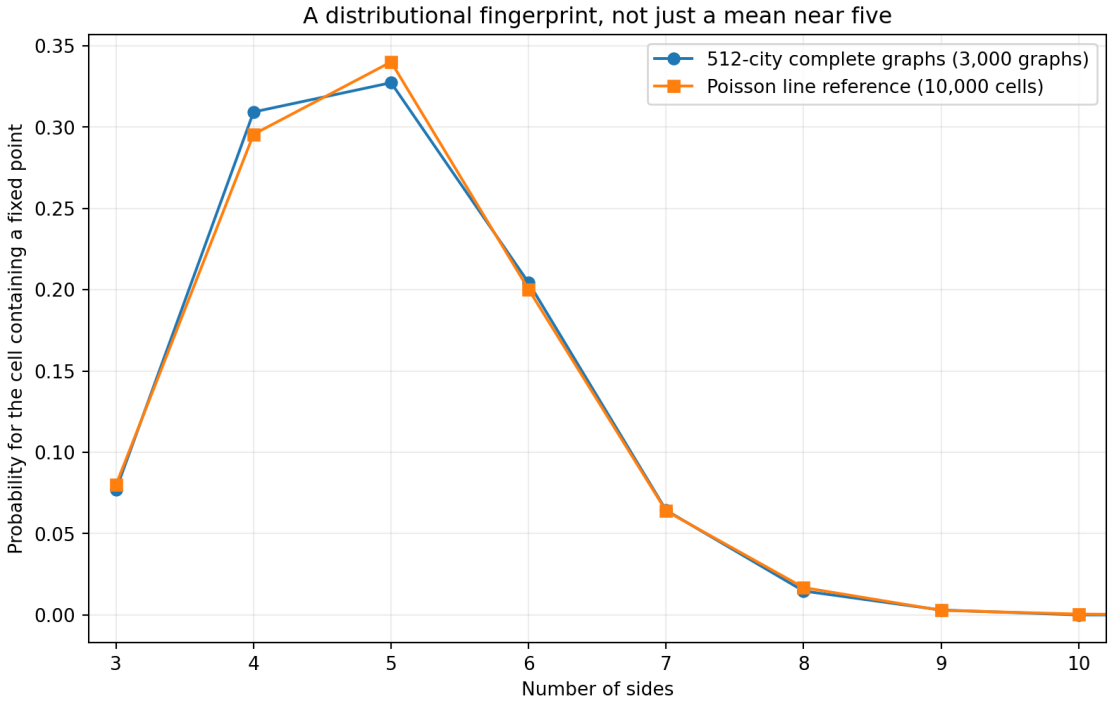


Figure 7: The complete side-count distribution at $n = 512$ resembles a separately simulated Poisson reference. Its largest mass is near five sides, not evidence that almost every cell is a pentagon.

7.5 Anisotropy and off-center behavior

Angular integration predicts normalized zero-cell mean side counts of 4.934802 at the disk center, approximately 4.934782 at $s = 0.65$ in the disk, and 4.909332 at the square center. The corresponding mean zero-cell areas at $L = 1$ are about 15.503139, 15.550643, and 15.522486.

The off-center finite graph converges more slowly: its measured area mean decreases from 25.259 at $n = 128$ to 19.134 at $n = 256$, still above the 15.551 reference. Its mean scaled clearance decreases from 1.130 to 1.064. These are genuine limitations of the finite-n approximation, not rows to discard because the fit is inconvenient.

8 Why the earlier angular-diversity test missed information

Normalize the angular field to a probability density $g(\theta) = \lambda(\theta)/L$ and define its undirected Fourier moments

$$m_j = \int_0^{\pi} e^{2ij\theta} g(\theta)\, \mathrm{d}\theta.$$

The absolutely convergent Fourier expansion of $|\sin t|$ gives

$$\boxed{v = \frac{L^2}{\pi}\left[1 - 2\sum_{j=1}^{\infty} \frac{|m_j|^2}{4j^2 - 1}\right].} \tag{16}$$

This is obtained by substituting

$$|\sin t| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{j\geq 1} \frac{\cos(2jt)}{4j^2 - 1}$$

into the crossing-intensity integral. It is valid for angular measures as well as smooth densities.

In particular, $v \le L^2/\pi$, with uniform angular distribution maximizing crossing intensity for fixed total line length. Consider three cases with the same L:

Directions	$1 - \lvert m_1 \rvert$	Crossing intensity v
All parallel	0	0
Equal horizontal and vertical	1	$L^2/4$
Uniform over all angles	1	L^2/π

The earlier diversity statistic $1 - |m_1|$ assigns the same maximal diversity to perpendicular-only lines and isotropic lines, despite their different intersection intensities. It measures only the second directional harmonic. Higher harmonics were invisible to that test.

For the square-center angular field, symmetry makes $m_1 = 0$, but $|m_2| \approx 0.21462$. Its efficiency $\pi v/L^2 \approx 0.99360$ is close to, but not equal to, one. The previous near-zero residual correlation with one chosen diversity statistic therefore did not establish that edge directions are irrelevant. Directional dependence of Poisson line cells is a classical subject; see, for example, [7].

9 What can and cannot be called an intrinsic scaling law

9.1 A consistency condition across city counts

For a nondegenerate bulk limit at fixed sampling domain, linear cell dimensions scale as n^{-2}, area as n^{-4}, and local crossing intensity as n^4. Suppose a proposed asymptotic monomial uses *that* crossing intensity q_n:

$$A/H \asymp (r/\sqrt{H})^{\beta}(Hq_n)^{\gamma}.$$

Since H stays of fixed order, consistency requires

$$-2\beta + 4\gamma = -4, \qquad \boxed{\beta - 2\gamma = 2.}$$

The attractive pair $(4/3, -1/4)$ gives $11/6$, not 2. Thus it cannot be the fixed-domain, high-n bulk law under that intensity interpretation. The old fitted pairs likewise fail this consistency check.

This is not a naive dimensional argument against the hull-normalized formula: that formula is invariant under overall geometric dilation for any exponents. It is an *across-n local-limit restriction.* Changing the definition of q changes the question; a radius-dependent count predictor is not automatically the theoretical q_n here.

The spacing expression $r\sqrt{d_2 d_3}$ has the correct total length degree two and the correct n^{-4} scale when all three distances are genuine local lengths. Nevertheless Section 2 shows that correct scaling does not provide universal accuracy.

9.2 Boundary layers and endpoint dependence

The proof concerns a fixed interior probe. At the boundary, long-range shared-city effects and the random convex hull are not negligible.

For the unit disk, a point at depth δ has an outward cap of area of order $\delta^{3/2}$. If $n\delta^{3/2} \to 0$, the probability of even one city in that cap tends to zero. Without such a city, the point is outside the

random convex hull. Thus the bulk face theory cannot be applied uniformly in the random-hull layer at depth much smaller than $n^{-2/3}$.

The deterministic fields $L(x)$ and $v(x)$ remain expectation identities, but expectations near a moving boundary can be influenced by rare configurations. A nonzero expected crossing density there is not a guarantee of a typical bounded probe cell.

9.3 No claim of exact finite-n independence

All incident edges at a city are highly dependent. They do not become globally independent as n grows. The Poisson statement is local: at the n^{-2} observation scale, the chance that the few visible edges share a city tends to zero. The $O(\log n/n)$ dependency estimate explains why residual dependence can still be visible at moderate n.

The word Poisson refers to the process of lines, not to an assertion that the induced crossing points are independent Poisson points. Crossings inherit dependence from their shared lines.

The model also does not say that every face becomes circular, that every face has five sides, or that a single deterministic area predictor can recover every cell. It predicts a nontrivial *distribution* of convex polygonal shapes.

10 Conclusions and reproducibility ledger

The earlier heuristic is now properly bounded in scope. Its nearest crossings can be topologically unrelated to the target face. The explicit adversary establishes a divergent error while preserving an elementary triangular target.

The stronger local object is an angular intensity field. Equations (1)–(4) turn a city-sampling distribution into exact finite-n expected length and crossing fields. They keep directional information that a scalar density or a single diversity moment loses.

A random-cell law replaces a fitted cell-area exponent. At the correct bulk scale, local complete-graph traces approach Poisson lines. This supplies exponential clearance, a containment functional, Gamma perimeter conditional on side count, and side-count predictions from classical stochastic geometry. Multiple independent observables support the same model.

Sampling is part of the theorem. The count-selected mean four and point-selected mean $\pi^2/2$ are different statements. Their relationship is controlled by an exact area bias and, more deeply, a convex volume product.

10.1 Claim ledger

Claim	Status
Nearest-crossing predictor has unbounded worst-case error	Analytic counterexample plus 41 numerical confirmations.
Finite-n segment and crossing expectation fields	Change-of-variables derivations; deterministic quadrature checks.
Bulk Poisson-line limit for complete graphs	Proof argument under stated iid/interior assumptions; no independent peer review or exhaustive novelty claim.
Zero-cell constants, Gamma law, area weighting and volume-product bounds	Classical results; derivations and primary references supplied.
Agreement of finite graphs with limiting distributions	Independent simulations; residual finite-size bias explicitly reported.
Convergence of all unbounded cell moments	Not established by the proof here.
Exact universal fractional exponents for individual faces	Not supported; earlier interpretation is withdrawn.

10.2 Archive contents

The `data/` directory contains every independent graph probe, including failed certification flags, seeds, and trial identifiers; the independent reference sample; all summary and distributional-test tables; the adversarial coordinates; and deterministic angular-field checks. The `code/` directory contains samplers, the local clipping algorithm, convex inball calculation, whole-graph verification, intensity quadrature, summarization, figure generation, and a full reproduction script. The `report/` directory contains this source and the final PDF. File hashes are listed in the manifest.

The complete-graph simulations were generated before the final presentation; there was no coefficient tuning to match the Poisson constants. The anisotropic and off-center cases, including their slower convergence, were retained. A reduced independent check can be obtained by running the sampler with a new trial-offset range, without changing any model parameter.

A Square-center crossing coefficient: an elementary calculation

By fourfold symmetry, write

$$\lambda(t) = \frac{1}{8\max(\cos t, \sin t)^3}, \qquad 0 \le t \le \pi/2.$$

On the half-quadrant $[0, \pi/4]$,

$$C_1 = \int \lambda(t)\cos t\,\mathrm{d}t = \frac{1}{8}, \qquad S_1 = \int \lambda(t)\sin t\,\mathrm{d}t = \frac{1}{16}.$$

On the entire quadrant, $C = S = 3/16$. The ordered sine integral over one half-quadrant is

$$J_1 = \int_0^{\pi/4} \lambda(t) \int_0^t \lambda(s)\sin(t-s)\,\mathrm{d}s\,\mathrm{d}t = \frac{1}{384}.$$

Reflection supplies the other half. The cross-half integral is $C_1^2 - S_1^2 = 3/256$. Hence the ordered whole-quadrant contribution is $J = 2J_1 + 3/256 = 13/768$. Combining the two quadrants in $[0, \pi)$,

$$v(0) = 2J + (C^2 + S^2) = \frac{26}{768} + \frac{18}{256} = \frac{5}{48}.$$

This calculation provides an independent exact target for the FFT angular quadrature.

B General zero-cell formulas for a nonuniform angular law

Set

$$\tau(\psi) = \int_0^\pi \lambda(\theta) |\sin(\theta - \psi)| \, \mathrm{d}\theta.$$

The probability that tu_ψ lies in the zero cell is $e^{-t\tau(\psi)}$. Polar integration yields

$$\mathbb{E}A_0 = \int_0^{2\pi} \frac{\mathrm{d}\psi}{\tau(\psi)^2}.$$

Campbell integration then gives

$$\mathbb{E}S_0 = v\mathbb{E}A_0, \qquad \mathbb{E}P_0 = L\mathbb{E}A_0.$$

These formulas explain why disk-center isotropy is convenient but unnecessary for a local theory. The domain constants in the archive were evaluated with 8,192 angular samples, with an exact isotropic calculation as a check. They are deterministic limiting-model references, not coefficients fitted to complete-graph observations.

References

[1] R. Arratia, L. Goldstein, and L. Gordon (1989). *Two Moments Suffice for Poisson Approximations: The Chen–Stein Method.* Annals of Probability 17(1), 9–25. doi:10.1214/aop/1176991491.

[2] M. Schulte and C. Thäle (2012). *The scaling limit of Poisson-driven order statistics with applications in geometric probability.* Stochastic Processes and their Applications 122(12), 4096–4120. arXiv:1201.5282.

[3] P. Calka (2003). *Precise formulas for the distributions of the principal geometric characteristics of the typical cells of a two-dimensional Poisson–Voronoi tessellation and a Poisson line process.* Advances in Applied Probability 35(3), 551–562. Author manuscript dated December 2002. Author-hosted manuscript.

[4] H. J. Hilhorst and P. Calka (2008). *Random line tessellations of the plane: statistical properties of many-sided cells.* arXiv:0802.1869, especially Section 1.2 for classical sidedness constants and references to Miles and Matheron. The large-sidedness asymptotics of that paper are not used as rigorous input here.

[5] R. Schneider (2009). *Weighted faces of Poisson hyperplane tessellations.* Advances in Applied Probability 41(3), 682–694. doi:10.1239/aap/1253281060. See Theorem 2, Equation (10), and the associated-zonoid volume-product argument in Section 3.

[6] K. J. Böröczky, E. Makai Jr., M. Meyer, and S. Reisner. *Volume product in the plane—lower estimates with stability.* Author-hosted manuscript, Theorem 13 for the centrally symmetric planar Mahler–Reisner inequality.

[7] N. Heerten, J. Krecklenberg, and C. Thäle (2022). *The proportion of triangles in a class of anisotropic Poisson line tessellations.* arXiv:2209.14121.